MIL-T-62497
8 February 1985

MILITARY SPECIFICATION

TRUCK, REFUSE COLLECTION: COMPACTION TYPE, ONE PERSON OPERATION, AUTOMATED SIDE-LOADING, 20 YARD, DUAL RIGHT AND LEFT HAND DRIVE, DIESEL ENGINE DRIVEN, 32,000 POUNDS GVW, 4x2, WITH TILT CAB, COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine driven, four-wheel, two wheel-drive, tilt cab, commercial, one-person operation refuse collection truck. The truck shall have dual right and left hand drive and a minimum gross vehicle weight (GVW) of 32,000 pounds. The minimum compaction volume of the body, excluding the hopper, shall be 20 cubic yards, and shall be side-loading and rear dumping. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GBS.l, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2320

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SPECIFICATIONS
FEDERAL

W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).

MILITARY

MIL-T-704	- Treatment and Painting of Material.
MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Number H-537.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117 - Standard Method of Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.

SAE Standards and Recommended Practices

J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (20-1000 MHz).
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six- and Eight-bolt Truck Transmission Mounted Power Take-offs (DoD adopted).
J816	- Engine Test Code - Spark Ignition and Diesel.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.7, the vehicles, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2), except for the following:

(a) The exterior of the complete body unit (see 3.5 through 3.5.4) shall be treated and painted in accordance with type A of MIL-T-704.
(b) The interior of the body shall be painted with rust-inhibiting primer and the manufacturer's standard color.
(c) The color of the exterior of the vehicle shall match color chip No. 13538, gloss yellow, of FED. STD. No. 595 for the Navy; color chip No. 16081, gloss gray, for the Air Force; and color chip No. 17875, gloss white, for all other services.

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(d) The color of vehicle identification markings shall be gloss black, color chip No. 17038, for the Navy; reflective white numerals and letters on a red background for the Air Force; and gloss black, color chip No. 17038, numerals and letters on a reflective white background on vehicle and on front and rear marking plates for all other services.
(e) Rubber seals and hydraulic hoses shall not be painted.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in the rear axle shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (ºF).

3.1.1.5 Coolant system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2-quart capacity. A low coolant level or high temperature coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel. On tilt cab models, a radiator servicing access door shall be provided to allow verification of coolant level.

3.1.1.6 Towing devices. Not less than two hooks, loops, pins, or a single center mounted towing eye for towing purposes, shall be furnished on the front of the vehicle.

3.1.1.7 Rear wheel splash and stone throw protection. The rear wheels shall have rubber mud flaps at the rear. Splash and stone throw protection sha1l be in accordance with SAE J682. Rubber mud flaps or fenders shall be provided to the rear of the front wheels and shall extend at least down to the centerline of the front axle.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The interior noise level during the packing and container hoisting cycles shall be within the permissible noise exposure levels of Occupational Safety and Health Standard 1910.95, based on an exposure duration of 8 hours per day (currently, 90 dBA). The drive-by vehicle exterior sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, equipment, and a full complement of fuel, lubricants, coolant, and hydraulic fluid.

3.2.5 Gross vehicle weight. The gross vehicle weight rating (GVWR) of the vehicle furnished shall equal the curb weight, the operator weight (computed at 175 pounds), and a payload equal to 650 pounds per cubic yard times the actual compaction volume of the body furnished; or shall equal 32,000 pounds, whichever is greater.

3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

3.2.8 Cab-to-axle dimension. The cab-to-axle dimension shall conform to that recommended by the refuse collection body manufacturer for conformance to this specification.

3.2.9 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

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3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle load to the specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall be capable of ascending a 1.0 percent grade at 50 miles per hour (mph). Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3). Gradeability requirements shall be met with the main transmission in direct drive.

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of recommended governed speed and shall provide a vehicle speed not greater than 4 mph.

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle on a 30 percent grade when the vehicle is loaded to its required GVW. The service brakes shall stop the vehicle, loaded to required GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE JB16 or SAE J1349. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for cooling.

3.4.1.2 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the maximum engine speed to the engine manufacturer's maximum recommended operating speed.

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The governor shall also provide an adjustable no-load speed setting which shall be maintained within 100 revolutions per minute (rpm) of the selected setting during the compaction cycle.

3.4.1.4 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. The control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

3.4.1.5 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. All heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperature (down to minus 60ºF) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab, and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) An immersion type engine oil heater, 300-watt minimum rating, with 170ºF to 195°F thermostat, shall be installed in the oil pan through any convenient opening.
(c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

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(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system with a 12-volt dc lighting system, and with not less than a 60-ampere alternator, shall be furnished. The engine starting equipment shall include an ether starting system or a glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, shall have a reservoir of not less than 12 fluid ounces, and shall be inoperative with the engine warm.

3.4.2.2 Lighting. The lighting system shall be a nominal 12 volts. All vehicle lights and reflectors shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Identification, sidemarker, and clearance lights shall be of the armored or recessed type with the exception of those lights installed on the roof of the truck cab. Shock and vibration reduction means shall be provided to prolong the filament life of all lights. Two stop lights, two tail lights, and two turn signal lights shall be furnished. These lights shall be installed on the rear at a height of not more than 60 inches from the ground. The lights shall be protected, either by recessed mounting below impact surfaces, or protected by guards equal in strength to 6-inch diameter standard weight pipe.

3.4.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on each steering column. The vehicle shall be provided with double-faced front signal units and with single-faced rear signal units, installed in accordance with SAE J588, except that the turn signals shall not be mounted on the engine compartment hood. Turn signal operating units shall have visible and audible type flash indicators.

3.4.2.4 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating shall be not less than 320 minutes and the total cold cranking ampere rating at 0ºF shall

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not be less than 1200 amperes, both measured in accordance with SAE J537. The batteries shall be of the maintenance-free type, having the maintenance-free characteristics listed in W-B-131.

3.4.2.5 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type air cleaner, oil capacity shall not be less than one quart.

3.4.3.2 Fuel tanks. Fuel tank capacity shall be not less than 43 gallons. When more than one tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. If furnished, vertical exhaust mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap. When specified (see 6.2), naturally aspirated engines shall be furnished with a spark arrester with an efficiency rating of not less than 80 percent when tested in accordance with SAE J350.

3.4.4.1 Exhaust system safety shield. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with ant part of the vehicle exhaust system.

3.4.5 Automatic or semiautomatic transmission. The vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall provide continuous drive. The transmission shall include a hydraulic torque convertor and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff opening(s). A hydraulic retarder shall not be used. If integral with the transmission, the hydraulic retarded shall be disconnected.

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3.4.5.1 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver. An air-actuated power takeoff shall be furnished.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard reinforced frame, or a heavy duty frame for the size vehicle furnished. The reinforced frame shall have frame reinforcement extending at least from the rear of the front suspension rear hanger bracket to the front of the rear spring. Reinforcement in the form of fishplating is not acceptable. Both the reinforced frame and the heavy duty frame shall have sufficient resisting bending moment to provide the structural strength required to at least equal the loads imposed with the vehicle loaded to provide the required GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to required GVW. When the suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Vehicles with a front axle rating of 12,000 pounds or less shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to its required GVW. Axle gear ratios shall provide the performance specified in 3.3.1. A single reduction rear axle shall be furnished.

3.4.9.1 Traction control. An automatic inter-wheel traction control shall be furnished on the drive axle differential. The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dua1 rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. The tire size and the rim size shall be the same for all wheels on the vehicle type wheels shall be furnished.

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3.4.10.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to required GVW. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of the heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.11 Brakes. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

3.4.11.1 Air brakes. Vehicles shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm)
(b) Air storage reservoir(s), with not less than 2,000 cubic inches total capacity. Each tank shall be equipped with a drain. Safety and check valves shall be provided between the compressor and the last reservoir tank
(c) Foot controls, suspended or treadle type, located for dual right and left hand drive
(d) Air control valves
(e) Air-pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Alcohol aspirator, with a transparent, unbreakable container
(i) Automatic moisture ejector
(j) Automatic slack adjusters
(k) When specified (see 6.2), in lieu of subparagraph (h) an air dryer shall be furnished. The air dryer shall be of the desiccant type.

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3.4.11.2 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished), and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system sha11 not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

3.4.12 Cab. A full-width tilt cab with tilting and locking mechanism shall be furnished. The cab doors shall be equipped with locks operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with an external, key-operated lock. Drip rails shall be installed above the cab doors. The cab shall have an upholstered individual driver's seat and an individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab. Interior lighting shall be provided. Fastenings and two pairs of seat belts shall be installed. Provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. The windshield wipers shall be of the multispeed type and shall be operated by either air or electric motor(s).

3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected by the body in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.16 Stowage space. Stowage space of sufficient size to accommodate hand tools, anti-skid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools and equipment may be furnished in the

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cab. When stowage space for these tools and equipment is located outside of the cab, the stowage space shall not be mounted on the body. The stowage space shall be weatherproof and shall be provided with provisions for locking with a padlock.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to the windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. Controls and operating mechanisms shall be located for both right and left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the left hand drive position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Dual sunvisors
(h) Driver's compartment ventilator other than window
(i) Tachometer.

3.4.19.1 Horn. Manufacturer's standard electric horn shall be furnished on each steering wheel.

3.4.19.2 Back-up alarm. When specified (see 6.2), the vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area.

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The convex portion shall have not less than 20 square inches of reflective area. The mirrors shall have not less than two supports.

3.4.21 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to assure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.5 Body. The body shall be of the side hopper loading, power compaction type. A hydraulic lift attachment, a loading hopper, and mechanisms that deliver refuse from the hopper into the body and compact the refuse shall be provided. Compacted refuse shall be discharged at the rear either by dumping, or by horizontal load ejection. The operations shall be actuated through selective controls and shall utilize power from the vehicle engine through the power takeoff. The installation of the body on the chassis shall be subject to the approval of the chassis manufacturer.

3.5.1 Construction. The body shall be of general welded construction, fabricated from steel sheets or plates reinforced with steel shapes to withstand continuous operation at maximum imposed loads without evidence of deformation or excessive wear of the body or its components. The interior of the body shall have a smooth, cleanable finish, shall be free of pockets and restrictions that might collect refuse, and shall be painted with a rust-inhibiting primer. The body shall load and retain all refuse material, including ashes and liquid-saturated garbage, without spillage or liquid leakage during both the refuse collection and the transport operations.

3.5.2 Hopper. The hopper shall be located directly behind the cab. The hopper shall have a capacity of not less than 3 cubic yards. At least one side of the hopper shall be equipped with an access door, to provide for container retrieval if necessary. The charging opening shall accommodate the emptying of not less than a 90-gallon container.

3.5.3 Compacting. The compacting plate shall clear the hopper by pushing its contents into the main cavity of the body. The plate shall not bypass refuse. Packing cycle shall be continuous and shall allow for hopper loading during all phases of compaction.

3.5.3.1 Compaction density. The compaction system shall be capable of compacting refuse to a density of not less than 650 pounds per cubic yard. This compaction density shall be considered normal service, such that in frequent compaction to this density, the stress levels required for long fatigue life of the vehicle are not exceeded.

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3.5.4 Container lift attachment. The hydraulic container lift attachment shall be capable of lifting not less than a 90-gallon container and 600 pounds. The attachment shall be mounted on the curbside of the vehicle, and the various size containers shall be accommodated by changing container grippers only. Grippers shall be designed to allow quick change-over without the use of special tools or equipment. The attachment shall have a horizontal side reach of not less than 36 inches. The lift cycle time shall be not more than 30 seconds. The attachment shall lift containers in a vertical position until the dumping action starts. When the container is in the full dump position, the container opening shall be completely within the confines of the hopper. The container lift attachment shall be designed to retract automatically to a position within the 96 inch vehicle width when not in use. Retraction shall be part of an automatic, uninterruptable cycle once the lift attachment is disengaged from the container, or shall occur automatically when the transmission is engaged.

3.5.4.1 Containers. When specified (see 6.2), containers shall be provided. The containers shall be stable and self-balancing when in the upright position either loaded or empty. Each container shall be capable of withstanding a compression force of not less than 200 pounds in the lifting and dumping process without permanent damage, deformation, or structural failure. Container capacity shall be not less than 80 gallons.

3.5.5 Refuse ejection. The rear ejection opening shall be full body width and height. The tailgate door shall be hinged at the top, or the side and shall, when open, provide an unrestricted opening for ejecting the refuse. The door shall be reinforced to prevent deformation which may cause leaks between door and body during packing operation. Latches shall be provided for holding the door securely in a locked position. Power-operated means shall be provided for swinging the tailgate away from the body to permit unobstructed ejection. Refuse shall either be discharged by dumping, or by horizontal ejection. When dumping is used, body props shall be provided.

3.5.6 Operating mechanisms. The mechanisms for emptying the hopper, compacting the refuse, lifting the container, opening the tailgate, and ejecting the refuse shall be hydraulically operated. The pump shall be a heavy duty, positive displacement type. Power to the pump shall be transmitted by a heavy duty, antifriction bearing type power takeoff through a heavy duty driveline. All mechanical operating mechanisms shall have an ultimate load capacity of not less

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than 2.0 times the maximum operating load. For ease of service, hydraulic valve assembly, including relief valve, shall be located so that it will be accessible at all times and under all load conditions.

3.5.6.1 Hydraulic system. Relief valves shall be provided in the hydraulic system to prevent excessive hydraulic pressure and shall be set to relieve hydraulic pressures in excess of that recommended by the manufacturer. A hydraulic T-fitting, or equivalent, for insertion of a pressure gage, shall be provided in the line beyond the main relief valve to facilitate checking the line pressure. All circulating fluid shall pass through a full-flow hydraulic fluid-filter. The filter element mesh shall be of a size to limit circulation of particles of 10-micron size or larger. The filter shall be located so that it will not contribute to pump cavitation and shall be accessible for servicing. Hydraulic lines, hoses, and fittings shall meet or exceed the requirements of Recommended Practices and Standards contained in the SAE Handbook. A reservoir shall be provided with sufficient flow volume to maintain the fluid temperature below the recommended limits and shall incorporate an oil level indicator. All valving shall be equipped to be stopped or reversed at any time during operation.

3.5.6.1.1 Hydraulic cylinders. All exposed hydraulic cylinders shall be fitted with seals to prevent entrance of dirt, moisture, and other foreign elements and to preclude leakage of hydraulic fluid. The exposed portions of hydraulic ram piston rods shall be chrome plated with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM Designation B-117. The connecting ends shall be treated in the best commercial practice.

3.5.6.1.2 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. A corrosion-resistant plate indicating the type of hydraulic fluid to be used shall be installed near the filler cap. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

3.5.6.2 Controls. Controls for the container lift, the packing mechanisms, and for refuse ejection or dumping shall be located in the right hand drive area of the cab. The controls shall be of the interrupted cycle or deadman type. Lift controls shall be lever actuated hydraulic. The speed of the engine shall be controlled automatically for proper operation of the body hydraulic system. A neutral interlock shall be installed to prevent advancing the throttle from the lift and ejection or dumping controls unless the vehicle automatic transmission is in neutral.

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3.5.7 Lubrication. All moving parts of the vehicle normally requiring lubrication shall be provided with a means for lubricating. Pressure grease fittings shall be of the hydraulic type with a surface check ball.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of the engine and brake systems; filling and charging of the battery; alinement of front wheels; inflation of all tires; complete lubrication of the chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All workmanship on the vehicle chassis-cab and on the body shall be in accordance with the highest standards of the vehicle and body manufacturing industries.

3.7.1 Metal fabrication. Metal used in the fabrication of the vehicle shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and cutting shall be done neatly and accurately in accordance with the standards specified in 3.7. Corners shall be square and true. All bends of a major character shall be made in a manner to insure uniformity of size and shape.

3.7.2 Bolted connections. Washers and lockwashers shall be provided and all bolts, nuts, and screws shall be tight. Bolt holes shall be accurately punched or drilled and shall be deburred.

3.7.3 Welding. The surfaces of parts to be welded shall be free from rust, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

4.3.2 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on the front and rear axles. The imposed loading on front and rear axles shall be computed using the curb weight, the operator weight at 175 pounds, and the payload equal to 650 pounds per cubic yard times the actual compaction volume of the body furnished, or 32,000 pounds, whichever is greater. Calculated imposed loads on front and rear axles will be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet contractual requirements.

4.3.3 Compaction body. When specified (see 6.2), the first production vehicle shall be tested using refuse to assure complete functional operation of the cycle of delivering refuse from the hopper into the body and discharging the refuse.

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4.3.4 Container lift. The container lift cycle shall be measured to determine conformance to 3.5.4. When specified (see 6.2), the first production vehicle shall be tested using the refuse container to assure complete functional operation of the lifting mechanism.

4.3.5 Road test. The vehicle shall be driven for a distance of not less than ten miles over class I (concrete, asphalt, or brick) roads at maximum allowable traffic speed limits but not less than 45 mph. The load shall be discharged and the vehicle returned to the contractor's plant.

4.3.6 Body treatment and painting. The certification regarding the body cleaning, treatment, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be presented to Government representatives for examination and approval.

4.3.7 Examination. When testing is completed, the vehicle shall be cleaned completely. The vehicle shall be examined for evidence of distortion, malfunction, and damage. The contractor shall be responsible for correcting defects disclosed during the operation and for taking corrective action to prevent recurrence of such defects in subsequently furnished vehicles.

4.3.8 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.9 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

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5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle is intended for nontactical use in loading, compacting, transporting, and dumping miscellaneous refuse generated at military installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).
(c) Rustproofing, if required (see 3.1.1.2).
(d) Tropical rustproofing, if required (see 3.1.1.2).
(e) Power plant heaters, if required (see 3.4.1.5).
(f) If battery heater is not required with power plant heaters (see 3.4.1.5).
(g) Spark arrester, if required (see 3.4.4).
(h) Disc type wheels, if required (see 3.4.10).
(i) Bias ply tires, if required (see 3.4.10.1).
(j) Desiccant type air dryer, if required (see 3.4.11.1(j)).
(k) Brake controls for use from a towing vehicle, if required (see 3.4.11.2).
(l) Back-up alarm, if required (see 3.4.19.2).
(m) Container(s), if required (see 3.5.4.1).
(n) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.5.6.1.2).
(o) Testing of refuse compaction and discharge cycle, if required (see 4.3.3).
(p) Testing of container lift attachment, if required (see 4.3.4).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, engine model number, and vehicle type. Computations should be based on vehicle operating under normal atmospheric pressure, in normal ambient air temperature, in still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Tables:

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Table 1	-	Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	-	Altitude Factor. 1.00
Table 3	-	Rolling Factor. 1.613
Table 4	-	Area Factor. 0.240
Table 5	-	Velocity Factor. 250.0
Table 6	-	Altitude Factor. 1.00
Table 7	-	Chassis Friction Horsepower. Use applicable power unit GVW (to nearest , higher, tabulated 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	-	Grade Factor. 0.75
Table 8A	-	Correction Factor. Not required.
Table 9	-	Road Factor. 0.0.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing activity:
Army - AT

Project No. 2320-1143

Review activity:
Navy - MC
Air Force - 84

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